

AC-ENTRY

Diagram illustrating AC Entry Protection components and connections:

- Input 1:** N_ACDC_1 connects to JACL1, then through STUD-M8 to JPE, and finally through F1 (FUSE-gG-690V) to LF1.
- Input 2:** N_ACDC_2 connects to JACL2, then through STUD-M8 to JPE, and finally through F2 (FUSE-gG-690V) to LF2.
- Input 3:** N_ACDC_3 connects to JACL3, then through STUD-M8 to JPE, and finally through F3 (FUSE-gG-690V) to LF3.

SURGE

Diagram illustrating Surge Protection components and connections:

- Input 4:** N_ACDC_4 connects to MOV3, then through S20K550 to LF1.
- Input 5:** N_ACDC_5 connects to MOV1, then through S20K550 to LF2.
- Input 6:** N_ACDC_6 connects to MOV2, then through S20K550 to LF3.

The diagram illustrates the power system configuration, divided into two main sections: AC-ENTRY and SURGE.

AC-ENTRY Section:

- AC Sources:** N_ACDC_1, N_ACDC_2, and N_ACDC_3 are connected to a common bus (P).
- Transformers:** JACL1, JACL2, and JACL3 are connected to the bus (P) and lead to three STUD-M8 units.
- PE Connection:** A PE (Protective Earth) connection is shown, leading to a common bus (P).
- Fuses:** F1, F2, and F3 are connected to the bus (P) and lead to three STUD-M8 units.

SURGE Section:

- AC Sources:** N_ACDC_4, N_ACDC_5, and N_ACDC_6 are connected to a common bus (P).
- Transformers:** GDT1, GDT2, and GDT3 are connected to the bus (P) and lead to three STUD-M8 units.
- PE Connection:** A PE (Protective Earth) connection is shown, leading to a common bus (P).
- Fuses:** F1, F2, and F3 are connected to the bus (P) and lead to three STUD-M8 units.

EMI-FILTER

CMC1

LF1 1 A1
AC1M 2 B1
LF2 3 A2
AC2M 4 B2
LF3 5 A3
AC3M 6 B3

CMC-3PH-2mH-SKU

CMC2

AC1M 1 A1
AC1D 2 B1
AC2M 3 A2
AC2D 4 B2
AC3M 5 A3
AC3D 6 B3

CMC-3PH-2mH-SKU

LDM2

AC2D 1 2 AC2

DM-22u-SKU

LDM3

AC3D 1 2 AC3

DM-22u-SKU

CX11

AC1M 1 2 AC2M

2.2uF

CX12

AC2M 1 2 AC3M

2.2uF

CX13

AC3M 1 2 AC1M

2.2uF

CX21

AC1 1 2 AC2

2.2uF

CX22

AC2 1 2 AC3

2.2uF

CX23

AC3 1 2 AC1

2.2uF

CY1

AC1 1 2 PE

4.7nF

CY2

AC2 1 2 PE

4.7nF

CY3

AC3 1 2 PE

4.7nF

LDM1

AC1D 1 2 AC1

DM-22u-SKU

Figure 1: Schematic diagram of the PRECHARGE section. The diagram shows two solenoid coils, KPRE1 and KPRE2, each represented by a yellow rectangle. KPRE1 is connected to V24 (pin 1), N_ACD_C_7 (pin 3), AC1 (pin 4), DGND (pin 5), AC1F (pin 6), and COIL_KPFE (pin 8). KPRE2 is connected to V24 (pin 1), RELAY_FB_KPFE (pin 3), AC2 (pin 4), N_ACD_C_7 (pin 5), AC2F (pin 6), and COIL_KPFE (pin 8). Both coils are labeled HF167F-80A-M. To the right, three relays are shown: RKFBP (10k) connected to V3P3 (pin 1) and RELAY_FB_KPFE (pin 2); RPFE1 connected to AC1 (pin 1) and AC1F (pin 2); and RPFE2 connected to AC2 (pin 1) and AC2F (pin 2). All relays are labeled CER-25W-AX.

DC-Modules – 30 kW module
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 File: acdc-INPUT-EMI.kicad_sch
Title: 30kW ACDC 1of6 INPUT-EMI

Size: User	Date: 2026-09-05	Rev: D.1
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